Semisupervised Semantic Segmentation of Remote Sensing Images With Consistency Self-Training

Jiahao Li, Bin Sun, *Member, IEEE*, Shutao Li, *Fellow, IEEE*, and Xudong Kang, *Senior Member, IEEE*

***Abstract*— Semisupervised semantic segmentation is an effective way to reduce the expensive manual annotation cost and take advantage of the unlabeled data for remote sensing (RS) image interpretation. Recent related research has mainly adopted two strategies: self-training and consistency regularization. Self-training tries to acquire accurate pseudo-labels to explicitly expand the train set. However, the existing methods cannot accurately identify false pseudo-labels, suffering from their negative impact on model optimization. The consistency regularization constrains the model by producing consistent predictions robust to the perturbations introduced in the sample or feature domain but requires a sufficient number of training data. Therefore, we propose a strategy for the semisupervised semantic segmentation of the RS images. The proposed model in the generative adversarial network (GAN) framework is optimized by consistency self-training, learning the distributions of both labeled and unlabeled data. The discriminator is optimized by accurate pixel-level training labels instead of the image-level ones, thereby assessing the confidence for the prediction of each pixel, which is then used to reweight the loss of the unlabeled data in self-training. The generator is optimized with the consistency constraint with respect to all random perturbations on the unlabeled data, which increases the sample diversity and prompts the model to learn the underlying distribution of the unlabeled data. Experimental results on the the large-scale and densely annotated Instance Segmentation in Aerial Images Dataset (iSAID) datasets and the International Society for Photogrammetry and Remote Sensing (ISPRS) datasets show that our framework outperforms several state-of-the-art semisupervised semantic segmentation methods.**

***Index Terms*— Consistency self-training, generative adversarial network (GAN), remote sensing (RS) image, semantic segmentation, semisupervised learning.**

I. Introduction

ADVANCES in earth observation technology have produced a large volume of high-resolution remote sensing (RS) images. Semantic segmentation is one of the critical techniques for interpreting such images to useful information, which is of vital importance to applications such as precision agriculture, urban planning, land-use surveillance, and traffic navigation.

Since the advent of the full convolutional neural network (FCN) [1], a large number of excellent studies [2]–[5] have emerged, which has effectively improved the performance of natural image segmentation. This trend has also promoted the development of RS image semantic segmentation [6]–[13]. Zhao *et al.* [11] first used a dual-tunnel convolutional neural network (CNN) architecture to capture spectral and spatial features. To enforce spatial consistency in the deeper layers of networks, a new pixel affinity matrix is defined. To take advantage of spectral information, spatial structure information, and semantic context-aware information in each pixel, Zhang *et al.* [10] extract representations of the diverse region and use the "multiscale summation"module to combine multiple scales and different level features. These methods based on supervised learning require a large number of pixel-level labels, which are labor-intensive and time-consuming to obtain manually. The average labeling time of a high-resolution urban scene image is 1.5 h [14]. Additionally, some hard cases may even require field verification to ensure accurate

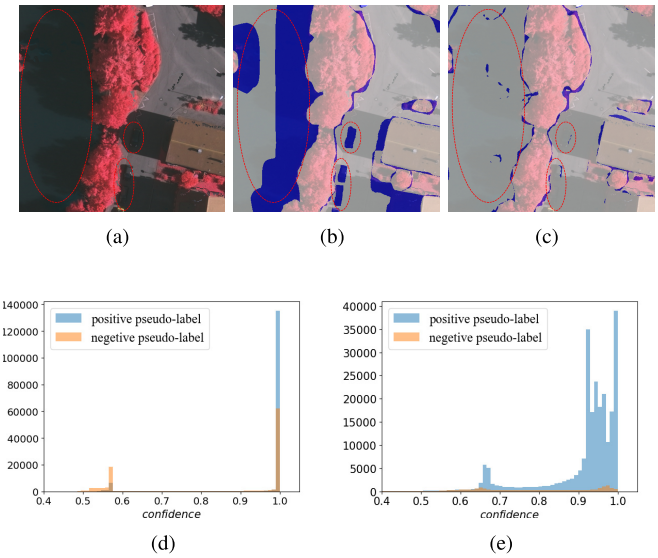

Fig. 1. (a) Local image on the Vaihingen test set. (b) False pseudo-label in Hung *et al.*'s method [20] (1/4 labeled data). (c) False pseudo-label in our method (1/4 labeled data). (d) Hung *et al.*'s [20] histogram of confidence assessment. (e) Our histogram of confidence assessment. The red dashed circle shows that our method makes an accurate confidence assessment for the area of shadows and small classes.

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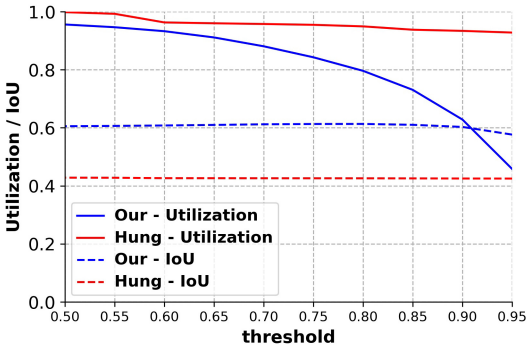

Fig. 2. IoU and utilization of pseudo-labels (all Vaihingen test set) with different T in our method (1/4 labeled data) and Hung *et al.*'s method [20] (1/4 labeled data). The Utilization refers to the area ratio of the pseudo-label to the entire area. The IoU refers to the area ratio of the pseudo-label to the positive prediction.

labeling due to poor light conditions, ambiguous spectral, low resolution, and so on. In [15] and [16], they use the weakly supervised method and the semisupervised fusion method to achieve competitive results. In [17], an effective unsupervised feature learning method was developed for multisource collaboration, and this work took full advantage of the unlabeled data, ensuring the completeness of information utilization and robustness of feature extraction even under small-training-sample. Therefore, the semisupervised learning has become an effective way to reduce the number of manually annotated samples and take advantage of the unlabeled data to improve the semantic segmentation performance.

Recently, a variety of different deep learning based semisupervised semantic segmentation methods have been proposed, which usually adopt two strategies: self-training [18]–[22] and consistency regularization [23]–[28].

Self-training usually uses predictions as pseudo-labels and minimizes the entropy between the predictions and pseudo-labels, thereby constraining the model to learn the classification characteristics of each pixel of the unlabeled data. Since false pseudo-labels used in self-training will deteriorate the performance, it is crucial to assess the confidence of predictions. In [18] and [19], the model prediction map is used to assess the confidence of prediction. However, the model prediction maps may be not accurate enough to serve as confidence assessment, because the model is sufficiently "confident" for its prediction of easier classes which often dominate self-training [29]. The model may be too confident for the misclassified pixel to avoid introducing too many false pseudo-labels. In [20] and [21], the generative adversarial network (GAN) [30] is introduced into semisupervised learning of semantic segmentation. The semantic segmentation network is used as the generator, while the role of the discriminator is to assess the confidence of predictions. However, the existing semisupervised semantic segmentation methods based on GAN cannot accurately identify positive predictions due to the problem of discriminator training strategy, so the accurate pseudo-labels used for self-training are very limited. Fig. 1(c) appears as a misjudgment in the regions of shadows and small class (e.g., car) using Hung *et al.*'s method [20]. And as shown in the red curve in Fig. 2, the gentle utilization curve implies that the existing discriminator training strategy cannot make pixel-level confidence assessment of predictions. The low intersection over union (IoU) curve reflects the limited access to accurate pseudo-labels.

Consistency regularization constraints model maintains consistency on predictions with random perturbations in the sample domain or feature domain. This strategy have been successfully used in classification tasks [31]–[33] due to *cluster assumption*, which states that the decision surface is located at low density area. Recently, semantic segmentation, as a high-level classification task, adopts consistency regularization. French *et al.* [34] observed that the low-density area does not exist on the category boundary, thereby proposing mask-based augmentation strategies. Ouali *et al.* [26] believed that the effectiveness of consistency regularization is limited by the characteristics of data distribution. Therefore, its effect on semantic segmentation is still to be explored.

In this work, two main difficulties for semisupervised learning are focused: accurate pseudo-label identification and prediction consistency constraint. We analyze in detail the problems of existing semisupervised semantic segmentation methods based on GAN and summarize the prerequisites for the success of consistency regularization in semisupervised semantic segmentation tasks, thereby proposing a semisupervised learning with consistency self-training framework. Specifically, derived from the idea of Mittal *et al.* [21], we use the semantic segmentation network as the generator to generate predictions and send them with random perturbations to the discriminator which assesses the confidence of these inputs. If they are considered as positive predictions, they will be used as pseudo-labels in self-training.

1) Different from the existing method using 0 and 1 image-level labels to train the discriminator [20], [21], we generate the discriminator labels through the pixel-level comparison between predictions and ground truth, and use these labels to supervise discriminator to assess the confidence of predictions.
2) According to the confidence assessment, we reweight the cross entropy loss of the unlabeled data during the training process of the generator, thereby effectively balancing the contributions of different quality pseudo-labels.
3) We introduce random perturbations (these are the same in [26]) in the generator's output layer [26] and fuse the predictions with all perturbations to increase the distribution difference of low-density regions in pseudo-labels, thereby constraining generator to maintain consistency of predictions with all perturbations.

The main contributions of this work are summarized as follows.

1) In accurate pseudo-label identification, the pixel-level comparison between prediction and ground truth provides local supervision for the discriminator to produce a more delicate confidence map instead of imitating the global distribution of the ground truth. Thus, the discriminator can assess the positive prediction more effectively.

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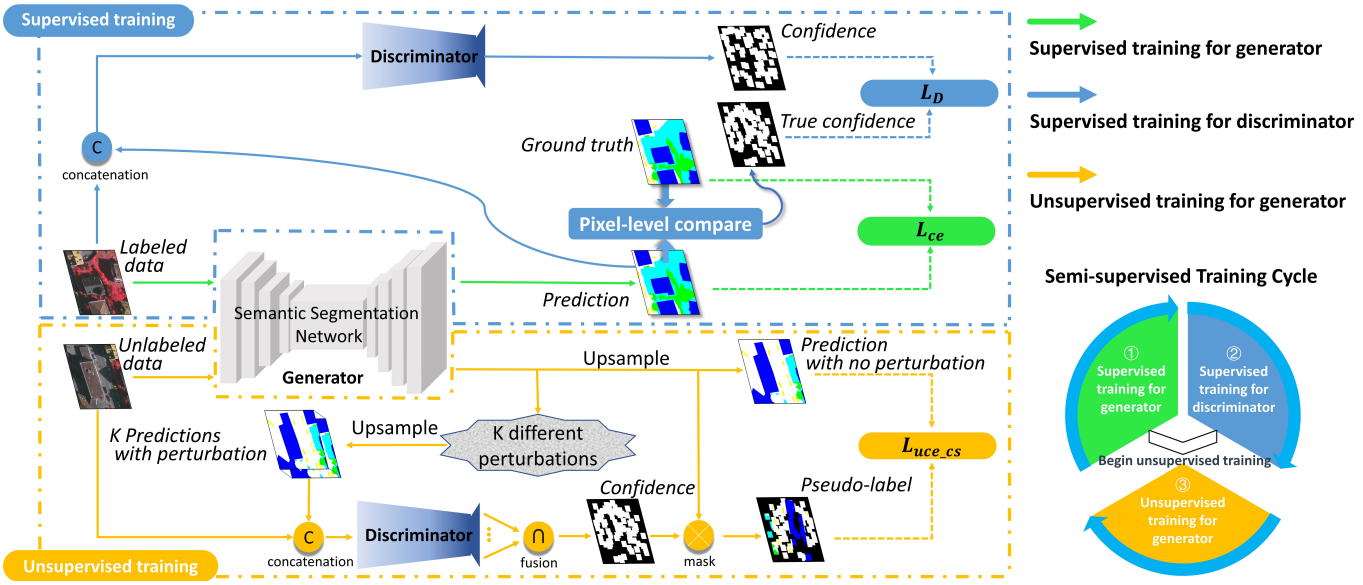

Fig. 3. Framework of the semisupervised learning with consistency self-training.

2) In prediction consistency constraint, the predicted labels with high confidence across all the perturbations are selected to make a more reliable all-to-one consistency constraint to improve the robustness of the generator instead of the conventional one-to-one consistency.

II. Related Work

A. Semisupervised Semantic Segmentation Based on GAN

Since the prediction maps of the segmentation model cannot characterize the confidence of predictions, additional models are required to undertake the task of assessing the confidence. Recently, the methods based on GAN have made progress in the field of semisupervised semantic segmentation. Luc *et al.* [35] introduced GAN to semantic segmentation for the first time. The semantic segmentation network is used as generator and added an adversarial loss on the basis of the traditional semantic segmentation network. Souly *et al.* [22] used FCN as a discriminator to produce confidence maps $(K+1)$ for each class (K) and a label for fake data to expand the labeled data. Hung *et al.* [20] replaced the generator being used to generate pseudo-labels and the discriminator being used to assess the confidence of predictions with segmentation network, thereby calculating the cross entropy loss between the predictions and pseudo-labels to realize online self-training. Mittal *et al.* [21] used discriminator to assess the image-level confidence of the entire prediction, and adopted the mean square error (MSE) loss to constrain the consistency of the distribution between predictions and ground truth. And they added a branch of teacher–student structure in the inference stage, ensuring high-level consistency of model predictions. However, existing GAN-based semisupervised learning labeled the discriminator training label of the predictions 0 and one of the Ground truth 1, which considers the image-level confidence of the entire prediction as to the one of each pixel and focus on the data distribution characteristics of manually annotated labels instead of positive predictions. Therefore, these discriminators cannot accurately assess the confidence of predictions. As shown in Fig. 2, the blue utilization curve implies that our method makes a pixel-level confidence assessment of predictions. The blue IoU curve implies that our method makes an accurate confidence assessment. Through the pixel-level comparison between predictions and ground truth to generate discriminator training labels, our method greatly improves the performance of predictions'confidence assessment.

B. Consistency Regularization

Originally, the purpose of regularization is to prevent the model from overfitting. The consistency regularization based on the *cluster assumption* hopes to learn from the distribution characteristics of unlabeled data in the low-density region, thereby avoiding the model from overfitting a small part of the labeled data. Researchers use data augmentation to increase the total amount of data to prevent model overfitting [36]–[38]. On this basis, Laine and Aila [23] and Sajjadi *et al.* [24] used the MSE loss to constrain the consistency in prediction of unlabeled data under different data augmentation. Tarvainen and Valpola [25] took the exponential moving average of model weights as the weights of the optimal model. The model is called "teacher model,"while the model of normal back-propagation to update the weights is called "student model."The purpose of consistency regularization is achieved through optimizing the prediction distribution loss of the above two models. Ouali *et al.* [26] believe that perturbations should be introduced in the feature domain instead of the sample domain because low-density regions exist in the feature domain of data. Although these studies continue to improve the consistency regularization methods, the constraining model to learn the distribution characteristics of the low-density region is not completely consistent with the goals of the classification task. The model cannot make accurate predictions for each unlabeled data, so the low-density region distribution characteristics of these unlabeled data may be negative. We introduce consistency regularization on the basis of self-training so

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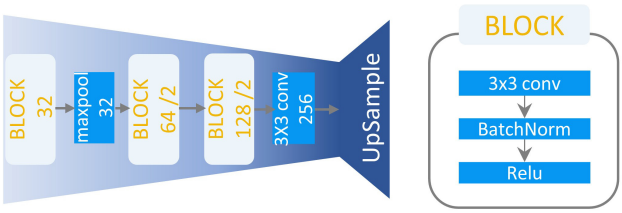

Fig. 4. Structure of the discriminator.

that the model can not only learn the category information of a single pixel of the image, but also learn distribution characteristics of each pixel in the low-density region.

C. Semisupervised Learning in RS

In the development of semisupervised semantic segmentation of RS images, some researchers replace pixel-level annotations with image-level annotations. Li *et al.* [39] used DualGAN to map the source images to the style of the target domain, thereby training cross-domain semantic segmentation network. Wei and Ji [40] trained a dual-branch encoder-decoder network with the proposal mask produced by a road label propagation algorithm and the auxiliary boundary prior information detected from images for road surface extraction. In addition to these methods under weakly-supervised constraints, some researchers directly use the unlabeled data to train model. In the methods based on self-training, Wu *et al.* [41] introduced spatial constraints in the model prediction map to correct and reassign the mistakenly classified pseudo-labels. In the methods based on consistency regularization, Kang *et al.* [42] consider the consistency in feature domain and sample domain. However, these methods are either based only on self-training or only on consistency regularization. In the RS community, there are few related research studies on semisupervised semantic segmentation.

III. Proposed Method

As shown in Fig. 3, the proposed method is based on the GAN. The semantic segmentation network serves as the generator, while the discriminator taking the labeled data and the predictions as the input produces the confidence assessing whether each pixel is accurately predicted. The entire training process is divided into two stages: supervised training and unsupervised training (respectively, the blue dashed line and the yellow dashed line in Fig. 3). In the supervised training stage, the two components of the GAN are trained with the labeled data in an alternative manner. First, the generator is trained with the labeled data while freezing the discriminator. Then, the discriminator is trained with the true confidence generated by comparing the prediction with the ground truth. In the unsupervised training stage, we introduce perturbations in the feature domain of the unlabeled data. We use the discriminator in inference mode to generate the corresponding confidence which is used to reweight the cross entropy loss of the unlabeled data to train the generator. In the test process, only the generator is used to make prediction for the input images.

A. Why Do We Need Consistency Self-Training

In the supervised learning, we define the entire dataset as $D^l = \{x^l, y^l\}$, where x^l and y^l represent the input image (data) and the ground truth, respectively, and train the model by optimizing the cross entropy loss. The cross entropy loss function in the semantic segmentation task is defined as follows:

$$L_{ce} = -\sum_{h,w,c} y^l_{(h,w,c)} \log S(x^l)_{(h,w,c)} \tag{1}$$

where $S(\cdot) \in H \times W \times C$, refers to the output of the segmentation network after the *softmax* function. H and W are the length and width of the input image, respectively. C is the total categories. y^l is the ground truth using one-hot encoding. In the semisupervised learning, we divide the entire dataset into labeled dataset $D^l = \{x^l, y^l\}$ and unlabeled dataset $D^u = \{x^u\}$. Since the unlabeled dataset cannot be trained with (1) due to the lack of the ground truth, semisupervised learning often needs to introduce additional loss to optimize the entropy of the unlabeled data.

Self-training utilizes the unlabeled dataset by optimizing the cross entropy loss between the predictions and the pseudo-labels

$$L_{\text{uce}} = -\sum_{h,w,c} y^u_{(h,w,c)} \log(S(x^u))_{(h,w,c)} \tag{2}$$

$$y^u = \text{onehot}(\text{argmax}(S(x^u))) \tag{3}$$

when optimizing the unsupervised cross entropy loss L_{uce}, the label y^u is derived from the segmentation network prediction $S(x^u)$ instead of manual annotation. Considering that the model will make negative predictions, not all predictions $S(x^u)$ are used to be considered as pseudo-label y^u. In the semisupervised learning based on GAN, the output of the discriminator is used as the criterion for selecting pseudo-label y^u. The selection strategy is defined as

$$y^u_{(h,w,c)} = \begin{cases} y^u, & D(\{S(x^u), x^u\})_{(h,w,c)} > T \\ -1, & D(\{S(x^u), x^u\})_{(h,w,c)} \le T \end{cases} \tag{4}$$

where $D(\cdot)$ refers to the output of the discriminator, $T \in (0, 1)$. We ignore the cross entropy loss of the part labeled -1. However, the existing semisupervised learning based on GAN often optimizes the cross entropy loss between predictions and 0 or ground truth and 1 when training the discriminator. Following this training strategy, the discriminator will simply assess the predictions based on whether it is close to the 0–1 distribution. It is difficult for the segmentation network to make a large-area prediction map that conforms to the 0–1 distribution for unlabeled data. Therefore, accurate pseudo-labels y^u that can be used to optimize the model will be very rare, thus wasting a lot of unlabeled data. Making full use of positive predictions in unlabeled data to optimize the unsupervised loss L_{uce} is the key to the success of self-training.

Consistency regularization believes that optimizing the relative entropy between predictions of unlabeled data with different perturbations can improve model performance. In machine learning, the relative entropy (i.e., *Kullback–Leibler* divergence) refers to the asymmetry measure of the difference

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between two kinds of distributions [43]. Considering that the low-density region exists in the embedding layer of the model, Ouali *et al.* [26] introduced perturbations in the feature domain and used the MSE loss to optimize the relative entropy between prediction with each perturbation and without perturbations

$$L_{\text{mse}} = ||S(x^u) - S_i(x^u)||^2 \tag{5}$$

where $S_i(\cdot)$ and $S(\cdot)$ refer to the output of the segmentation networks with ith perturbation and without perturbations in the feature domain, respectively. However, in semantic segmentation, the constraint target of the relative entropy loss is not consistent with that of the cross entropy loss. The latter is to make positive decision surface. The former is to learn the distribution characteristics of the low-density regions. Consistency regularization based on the *clustering assumption* requires the model to make positive decision surface (i.e., positive predictions). Otherwise, the relative entropy loss between predictions with random perturbations will make the model converge toward the negative distribution. In semisupervised learning, a small amount of labeled data may cause the model to be unable to make the positive decision surface in the entire dataset (include the unlabeled dataset). Then, optimizing the relative entropy loss of the unlabeled dataset will make the model performance worse.

Inspired by Ouali *et al.* [26], our semisupervised learning with consistency self-training framework introduces random perturbations in the feature domain of the segmentation network and optimizes the relative entropy between prediction with all perturbations and without perturbations by using the cross entropy loss. Specifically, we use the discriminator to assess the confidence of each prediction after introducing perturbations and fuse all predictions (K) for self-training. According to (2), the model is optimized by the cross entropy loss of the unlabeled data with all perturbations instead of each one.

B. Pseudo-Label Identification

Since accurate pseudo-label identification requires the discriminator to make an accurate confidence assessment of predictions, the research on the training method of the discriminator is particularly important.

As shown by the blue arrow line in Fig. 3 (i.e., supervised training for the discriminator), we only use the labeled dataset to train the discriminator. Since the input of the discriminator comes from the output of the segmentation network, we alternately freeze the discriminator and the segmentation network. The input x^l and the prediction $S(x^l)$ of the segmentation network are concatenated as the input of the discriminator. The result of the pixel-level comparison between the prediction and the ground truth is used as the training label of the discriminator. The comparison rules are defined as follows:

$$y^D_{(h,w,c)} = \begin{cases} 1, & \text{onehot}(S(x^l))_{(h,w,c)} = \text{onehot}(y^l)_{(h,w,c)} \\ 0, & \text{onehot}(S(x^l))_{(h,w,c)} \neq \text{onehot}(y^l)_{(h,w,c)} \end{cases} \tag{6}$$

where y^D refers to the discriminator training label, namely true confidence. We train the discriminator by optimizing the cross entropy between the confidence $D(\cdot)$ and the true confidence y^D. The loss function of the discriminator is defined as

$$L_D = -\sum_{h,w,c} y^D_{(h,w,c)} \log(D(\{S(x^l), x^l\})_{(h,w,c)}. \tag{7}$$

Therefore, our discriminator focuses on the positive prediction area and the negative prediction area, instead of regarding the predictions as fake and the ground truth as true. With the alternate training of the segmentation network and the discriminator, the discriminator can learn the segmentation characteristics of the segmentation network more fully, so as to accurately identify positive predictions. This is essentially different from the existing discriminator training strategy [20], [21]. The former focuses on the segmentation quality and regards the positive prediction area as true. The latter focuses on the quality of manual labeling and regards the ground truth as true. It is worth noting that although we only use the labeled data to train the discriminator, a small amount of training data can also train an excellent model due to the simple structure of the discriminator (see in Fig. 4).

C. Prediction Consistency Constraint

As shown by the yellow arrow line in Fig. 3 (i.e., unsupervised training for the generator), the discriminator are used in the inference mode. Inspired by Ouali *et al.* [26], we introduce K perturbations in the feature domain of the segmentation network. After that, we get the $K+1$ confidence of predictions (with K perturbations and without perturbations). According to (8), we fuse K confidence with perturbations, thereby acquiring a pseudo-label y^u with all perturbations

$$y^u_{(h,w,c)} = \begin{cases} y^u, & D(\{S_k(x^u), x^u\})_{(h,w,c)} > T \\ -1, & D(\{S_k(x^u), x^u\})_{(h,w,c)} \leq T \end{cases} \tag{8}$$

where $k \in [1, K]$, S_k represents predictions with perturbations. K is 7. The seven perturbations we introduced are defined as three types: feature based perturbations, prediction based perturbations and random perturbations, which have been verified effective. For more details, please refer to [26].

In the unsupervised training stage, the generator is optimized by the cross entropy between the pseudo-label with all perturbations and the prediction with no perturbations, instead of the relative entropy between the predictions with each perturbation and without perturbations. Therefore, adopting this method allows the segmentation network to maintain the consistency of the predictions with all perturbations, rather than the one with a single perturbation. Since pseudo-labels will be very diverse due to the randomness of perturbations, the labeled dataset will be expanded. In addition, the fused pseudo-label that has masked negative prediction area can avoid the problem of model deterioration caused by negative decision surface. To make the model more robust, we reweight the cross entropy loss in unsupervised training stage, which is defined as (9), according to the confidence assessment of the discriminator

$$L_{\text{uce_cs}} = -\sum_{h,w,c} w_{(h,w,c)} y^u_{(h,w,c)} \log(S(x^u))_{(h,w,c)} \tag{9}$$

$$w_{(h,w,c)} = D(\{S(x^u), x^u\})_{(h,w,c)} \tag{10}$$

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TABLE I
IN 1/4 LABELED DATA, THE IMPACT OF DIFFERENT T ON THE VAIHINGEN TEST SET

Labeled Data	T	OA	F1	Mean IoU
1/4	0.95	83.29	80.03	67.37
	0.9	83.43	80.43	67.86
	0.85	**83.68**	**80.59**	**68.09**
	0.8	83.48	80.31	67.74
	0.75	83.50	80.26	67.67
	0.7	83.33	79.92	67.25
	0.65	83.09	80.06	67.35
	0.6	83.59	80.39	67.85
	0.55	83.33	79.93	67.26
	0.5	83.59	80.39	67.89

TABLE II
IN 1/8 LABELED DATA, THE IMPACT OF DIFFERENT T ON THE VAIHINGEN TEST SET

Labeled Data	T	OA	F1	Mean IoU
1/8	0.95	81.22	77.34	63.82
	0.9	80.35	75.86	62.02
	0.85	81.18	76.97	63.43
	0.8	80.74	75.81	62.13
	0.75	81.09	77.25	63.66
	0.7	80.87	76.37	62.71
	0.65	81.04	76.91	63.33
	0.6	81.62	77.40	64.00
	0.55	**81.82**	**77.98**	**64.68**
	0.5	81.50	77.12	63.69

where $w_{(h,w,c)}$ refers to the confidence on the pixel (h, w, c). By optimizing the unsupervised cross entropy loss $L_{\text{uce_cs}}$, the model can learn the distribution characteristics of the unlabeled data, and balance the contribution of each area in the pseudo-labels according to the confidence assessment $w_{(h,w,c)}$.

In our semisupervised learning with consistency self-training framework, an dynamic generator loss is defined to start our semisupervised training cycle in an asymptotical way

$$L_S = L_{\text{ce}} + a(t)L_{\text{uce_cs}} \tag{11}$$

where $a(t)$ is the weight of unsupervised learning loss which varies with the iteration number t. The weight is calculated as follows:

$$a(t) = \begin{cases} 0, & t < T_s \\ \dfrac{t - T_s}{T_f - T_s}\alpha_f, & T_s \le t < T_f \\ \alpha_f, & t \ge T_f \end{cases} \tag{12}$$

where α_f is the full weight of the unsupervised learning loss. α_f affects training stability. If it is too large, the training will be unstable. Usually, we set α_f to 0.3. The unsupervised learning is not started until the T_sth iteration. Then its weight increases linear and reaches the full weight α_f at the T_fth iteration. T_s and T_f are usually set as one-third and two-thirds of the maximal number of iterations in our experiments.

IV. EXPERIMENTS

A. Datasets and Experimental Setup

1) Datasets: We evaluate the performance of our method on the large-scale and densely annotated Instance Segmentation in Aerial Images Dataset (iSAID) dataset [44] and Remote Sensing [International Society for Photogrammetry and Remote Sensing (ISPRS)] *Vaihingen*[1] and *Potsdam*[2] dataset.

The images of *iSAID* are the same as the a large-scale Dataset for Object Detection in Aerial images (DOTA) dataset [45]. The dataset consists of 2806 high-resolution images. The size of the images ranges from $12\,029 \times 5014$ to 455×387. The dataset contains 655 451 object instances with 15 categories. The categories are ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, and harbor. In the following experiment, the training set contains 1411 images, and the test set contains 458 images. We randomly crop the image to 512×512.

Vaihingen is a relatively small village with many freestanding buildings and small multistory buildings. The Vaihingen dataset consists of 33 images. The size of each image ranges from 1996×1995 pixels to 3816×2550 pixels. In the following experiment, 16 images were used for training and 17 images were used for testing. Taking into account the hardware limitations, we cut all data into a size of 512×512 pixels, and the overlap size is 128 pixels.

Potsdam is a typical historic city with huge buildings, narrow streets, and densely populated settlements. The Potsdam dataset contains 38 images with a size of 6000×6000 pixels, of which 24 are used for training and 14 are used for testing. We cut the Potsdam dataset in the same way as the Vaihingen dataset.

2) Experimental Setup: We use FCN [1] with ResNet-50 [46] and UNet [47] as the segmentation network in our method, respectively, to compare with existing excellent methods in the semisupervised and supervised learning. We use FCN with ResNet-18 and ResNet-50, respectively, to conduct ablation experiments to observe the improvement caused by each module. The labeled data mentioned in all experiments refer to the ratio of the labeled data to the train set. Unless otherwise specified, the unlabeled data all come from the train set. All experiments run on two NVIDIA GTX2080ti GPUs. We set the batch size of each GPU to four, including two labeled images and two unlabeled images. One epoch refers to an entire iteration of all unlabeled dataset. In the training process, we use stochastic gradient descent (SGD) with weight decay of 5e-4 and momentum of 0.9 [48] to optimize the segmentation network, and *Adaptive Moment Estimation* (Adam) to optimize the discriminator network. Data augmentation include random resizing, random cropping, and random flipping. We set the training epoch to 150, T_f to 100 epoch and α_f to 0.3. For the *Vaihingen* and *Potsdam* datasets, T_s is 50 epoch. For the *iSAID* dataset, T_s is 10 epoch.

[1]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

[2]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html

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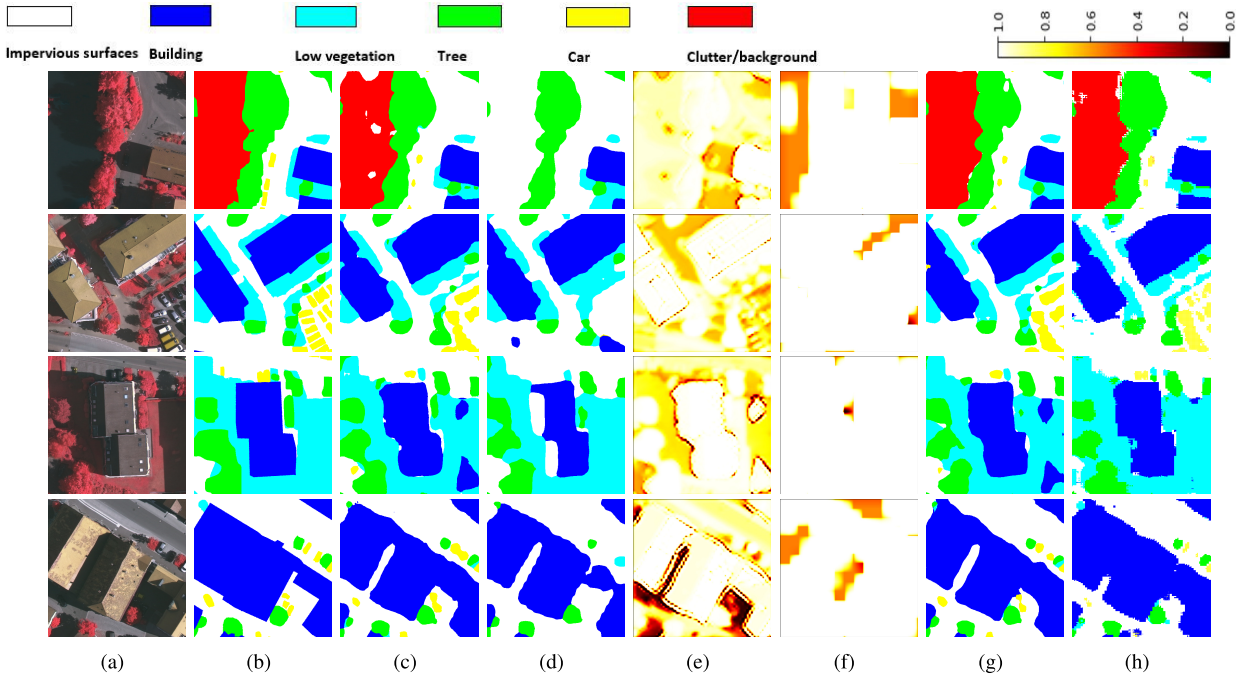

Fig. 5. Comparison between our method and Hung *et al.*'s method [20] on Vaihingen test set (1/4 labeled data). (a) Input image. (b) Ground truth. (c) Ours method. (d) Hung *et al.*'s method [20]. (e) Ours confidence assessment. (f) Hung *et al.*'s [20] confidence assessment. (g) S4GAN's method. (h) CCT's method.

TABLE III
COMPARED WITH SEMISUPERVISED METHODS ON THE VAIHINGEN TEST SET

Labeled Data	Method*	F1					OA	Mean IoU
		Surface	Building	Low-Veg	Tree	Car		
1/8	baseline	80.86	83.71	71.06	82.26	44.77	79.50	58.74
	Hung *et al.*[20]	82.85	**87.17**	**74.95**	**84.14**	0.02	**82.19**	56.37
	s4GAN[21]	**83.09**	84.97	73.31	83.06	60.28	81.16	63.40
	CCT[26]	81.30	84.66	69.64	81.72	40.71	79.54	57.99
	Ours	83.03	86.33	73.57	83.72	**63.24**	81.82	**64.68**
1/4	baseline	83.80	87.05	74.05	81.82	58.21	81.69	63.66
	Hung *et al.*[20]	84.78	88.50	74.91	82.91	0.07	82.66	56.74
	s4GAN[21]	85.44	88.78	75.61	82.91	66.86	83.30	67.25
	CCT[26]	85.16	**89.84**	**76.75**	82.25	54.14	83.54	64.99
	Ours	**85.67**	88.96	76.11	**83.48**	**68.69**	**83.68**	**68.29**

* the FCN with ResNet-50 is used as baseline and for all the SSL methods

TABLE IV
COMPARED WITH SEMISUPERVISED METHODS ON THE POTSDAM TEST SET

Labeled Data	Method*	F1					OA	Mean IoU
		Surface	Building	Low-Veg	Tree	Car		
1/8	baseline	87.26	91.29	80.76	80.55	73.03	85.07	70.82
	Hung *et al.*[20]	-	-	-	-	-	-	-
	s4GAN[21]	88.64	92.01	81.36	81.93	84.09	86.49	75.62
	CCT[26]	87.31	91.12	80.32	81.62	79.36	85.22	72.60
	Ours	**89.18**	**92.59**	**82.32**	**83.39**	**86.63**	**87.07**	**76.92**
1/4	baseline	88.82	92.72	82.49	82.09	79.31	86.68	74.37
	Hung *et al.*[20]	-	-	-	-	-	-	-
	s4GAN[21]	88.81	92.92	81.20	80.78	85.09	**87.42**	76.62
	CCT[26]	88.33	93.00	81.26	**83.04**	82.34	86.46	75.09
	Ours	**89.67**	**93.83**	**82.75**	82.22	**86.92**	87.26	**77.39**

* the FCN with ResNet-50 is used as baseline and for all the SSL methods

We use the following evaluation metrics: overall accuracy (OA), F1, and Mean IoU.

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{N} \tag{13}$$

$$F_1 = (1+\beta)\frac{\mathrm{precision} \cdot \mathrm{recall}}{\beta^2 \cdot (\mathrm{precision} + \mathrm{recall})}, \quad \beta = 1 \tag{14}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{FN} + \mathrm{FP} + \mathrm{TP}} \tag{15}$$

where TP, TN, FP, FN, and N are the numbers of true positive, true negative, false positive, false negative, and the total pixel number of the whole image, respectively.

B. Comparing With Semisupervised Works

In Fig. 2, we analyzed the impact of different T on the pseudo-labels. In Hung *et al.*'s method [20], the utilization curve has almost no fluctuations as the T increases. This shows that the method makes image-level confidence assessment, due to the prediction confidence closing to 1 for almost all pixels. The IoU between the pseudo-label and the positive prediction in our method increases by almost 2%.

We also evaluated the impact of different T on the performance in our method, as shown in Tables I and II. We observe that the T can be set more strictly to avoid the negative impact of false pseudo-label as the labeled data being more. The lower T can lead to higher performance if the labeled data is less. Therefore, appropriately relaxing the T can greatly improve the model performance. We set the T as 0.85 in 1/4 labeled data and as 0.55 in 1/8 labeled data.

Table III shows the comparison between our method and other semisupervised methods on the Vaihingen test set. In 1/8 labeled data, our method improves the OA, mean F1, and mean IoU by 2.32%, 5.44%, and 5.94% compared with baseline.[3] In 1/4 labeled data, our method improves OA, mean F1,

[3]FCN with ResNet-50 using supervised learning.

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TABLE V
COMPARED WITH SEMISUPERVISED METHODS ON THE iSAID TEST SET

Labeled Data	Method *	F1																OA	Mean IoU
		BG	Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SBF	Plane	Harbor		
1/8	baseline	96.68	25.37	nan	42.43	76.27	11.22	44.57	nan	63.50	17.94	nan	17.40	56.47	55.14	33.86	17.50	93.39	25.29
	Hung *et al.*[20]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	s4GAN[21]	97.26	15.59	17.21	69.12	84.72	**41.12**	51.99	nan	**72.29**	24.98	nan	25.56	56.33	54.81	34.02	34.73	93.96	31.49
	CCT[26]	**98.05**	**36.32**	**23.86**	19.30	86.38	40.85	28.64	nan	63.00	30.38	nan	0.27	59.95	**61.83**	25.04	**46.39**	93.62	28.57
	Ours	97.33	25.28	18.43	**72.45**	**86.49**	28.27	**53.38**	nan	71.93	**42.69**	nan	**50.47**	**68.39**	23.08	**62.18**	30.93	**94.60**	**34.42**
1/4	baseline	96.85	44.41	nan	58.73	80.10	61.32	37.17	nan	59.20	33.79	nan	54.43	62.27	47.45	38.14	35.40	93.75	32.44
	Hung *et al.*[20]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	s4GAN[21]	97.48	6.55	0.60	**74.39**	86.70	71.50	50.10	nan	68.98	41.79	nan	35.16	**74.85**	**79.56**	23.38	32.16	94.67	36.39
	CCT[26]	96.71	41.35	**20.92**	59.23	**93.60**	55.96	**53.77**	nan	72.61	36.92	nan	23.01	64.84	59.88	37.43	**49.46**	94.77	37.00
	Ours	**97.77**	**49.47**	20.61	67.17	92.32	**77.72**	40.77	nan	**74.36**	**51.85**	nan	**72.98**	48.60	60.08	**48.40**	43.51	**95.57**	**40.74**

* the FCN with ResNet-50 is used as baseline and for all the SSL methods

TABLE VI
COMPARED WITH SUPERVISED METHODS ON THE VAIHINGEN TEST SET

	Method	Classes IoU					Mean IoU	F1	OA
		Surface	Building	Low-Veg	Tree	Car			
SL	FCN[1]	77.68	82.33	63.94	74.35	51.83	70.02	81.46	87.11
	UNet[47]	79.02	84.46	65.23	75.13	57.07	72.18	84.44	87.14
	PSPNet[49]	78.90	84.26	65.34	*75.14*	56.18	71.96	84.15	87.17
	FPN[50]	77.65	82.72	64.34	74.43	52.11	70.25	82.94	86.70
	DeepLab_v3[51]	79.23	83.70	64.88	*75.14*	57.54	72.10	84.42	*87.32*
	DANet[52]	78.55	82.10	64.18	74.80	56.36	71.20	83.95	87.07
	BiseNet_v2[53]	77.70	78.78	63.23	74.98	53.24	69.58	83.57	86.88
SSL	Ours(FCN)	*79.32*	*85.59*	*65.96*	**75.33**	*61.72*	*73.59*	*84.49*	86.72
	Ours(UNet)	**80.27**	**86.38**	**66.60**	75.06	**64.27**	**74.52**	**85.62**	**87.44**

TABLE VII
COMPARED WITH SUPERVISED METHODS ON THE POTSDAM TEST SET

	Method	Classes IoU					Mean IoU	F1	OA
		Surface	Building	Low-Veg	Tree	Car			
SL	FCN[1]	81.67	88.99	71.24	72.80	79.91	78.92	85.75	85.64
	UNet[47]	82.58	90.08	72.58	**74.22**	81.38	80.17	88.67	87.76
	PSPNet[49]	82.71	90.21	72.62	**74.22**	81.11	80.18	*89.45*	88.78
	FPN[50]	81.63	89.22	71.37	73.05	79.09	78.87	88.72	88.27
	DeepLab_v3[51]	82.81	89.95	72.22	74.11	**82.16**	80.25	**89.66**	88.97
	DANet[52]	82.27	89.15	71.77	73.70	81.72	79.72	89.37	88.82
	BiseNet_v2[53]	81.17	87.61	71.75	73.60	81.41	79.11	88.78	88.18
SSL	Ours(FCN)	*85.13*	*91.40*	*74.03*	73.38	79.06	*80.60*	89.10	**89.73**
	Ours(UNet)	**86.06**	**92.76**	**74.66**	73.68	*81.87*	**81.81**	89.33	*89.47*

and mean IoU by 1.99%, 3.60%, and 4.63% compared with baseline. Our method has more obvious improvement in less labeled data. In addition, our method in 1/8 labeled data is better than baseline in 1/4 labeled data. We also analyzed the differences between our method and existing excellent semisupervised learning methods. In 1/8 labeled data, our method is lower than Hung *et al.*'s method [20] in OA, while much higher than theirs in Mean F1 and Mean IoU. Even if the labeled data increases, their method has a terrible performance on cars, which shows that the model cannot learn the positive knowledge suffering from a large volume of false pseudo-label. In cross-consistency training (CCT) representing consistency regularization, the method performs better as the labeled data increases. However. in less labeled data the method makes performance worse, and even worse than baseline. Our method performs well in 1/8 and 1/4 labeled data especially on cars, which shows that it can avoid negative effect caused by negative predictions and less labeled data to better learn the distribution characteristics in unlabeled data. As shown in Fig. 5, the local predictions show that our method makes pixel-level confidence assessment to identify positive prediction.

Table IV shows the comparison between our method and other semisupervised methods on the Potsdam test set. Our method improves the Mean IoU by 6.12% (1/8 labeled data) and 3.02% (1/4 labeled data) compared with baseline. Consistency regularization (CCT) plays a positive role due to the more amount of train data in the Potsdam dataset, while worse than our method. Our improvement mainly reflects on cars, which is consistent with the Vaihingen dataset.

Table V shows the comparison between our method and other semisupervised methods on the *iSAID* test set. In 1/8 labeled data, the CCT only improves the Mean IoU by 3.28%. Our method improves by 9.13%. In addition, for baseball diamond, ground track field, swimming pool, and plane, the CCT performs worse than baseline. Our method performs better on these classes with a small amount of labeled data. In 1/4 labeled data, our method improves the Mean IoU by 8.3%. Compared with the baseline in 1/4 labeled data, our method in 1/8 labeled data improves the Mean IoU by 1.98%. It can be seen that our method also performs very well on the large-scale dataset. This is because our semisupervised learning framework with consistency self-training uses an effective self-training to constrain the model to optimize in the direction of the positive decision surface, thereby ensuring consistency and improving model performance.

C. Comparing With Supervised Works

We compare with the existing excellent supervised learning methods on the Vaihingen test set in Table VI. Our method uses all train sets as labeled dataset and all test sets as unlabeled dataset for training. In Table VI, our improvement on impervious surfaces, buildings, low vegetation, trees, and cars are 0.09%–1.67% and 1.13%–6.81%, 0.62%–2.73%, 0.19%–0.98%, 4.18%–9.89%, respectively. Our improvement on car perform most significantly. Taking into account that FCN with ResNet-50 and UNet being used for segmentation, our method still performs better than the excellent segmentation network. Fig. 6 shows the whole image segmentation results of DeepLab_v3 [51], pyramid scene parsing network (PSPNet) [49], and our method (FCN) on Vaihingen test set.

We compare with the existing excellent supervised learning methods on the Potsdam test set in Table VII. In Table VII, our improvement mainly reflects on impervious surfaces,

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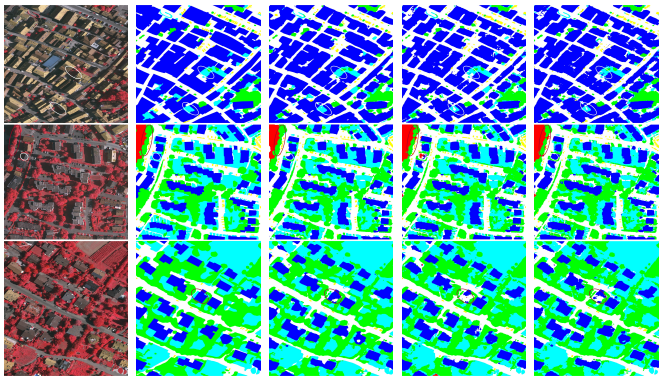
TABLE VIII
IMPROVEMENT COMPARISON OF EACH MODULE ON THE VAIHINGEN TEST SET

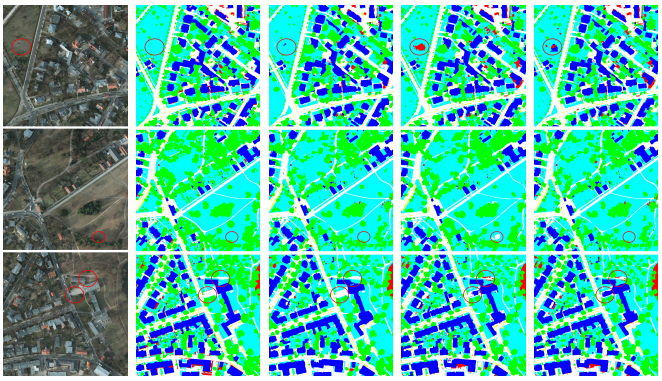
Backbone	Labeled Data	Method *	F1					OA	Mean IoU
			Surface	Building	Low-Veg	Tree	Car		
ResNet-18	1/8	baseline	78.54	80.42	64.86	80.09	0.50	76.23	49.35
		$+L_D$	80.48	82.38	70.23	**82.18**	46.41	78.89	58.30
		$+L_D + L_{uce}$	80.86	82.51	**71.16**	81.48	47.41	79.01	58.64
		$+L_D + L_{uce_cs}$	**81.64**	**83.96**	70.80	81.72	**53.16**	**79.67**	**60.29**
	1/4	baseline	81.25	84.25	70.69	79.25	24.58	78.78	55.11
		$+L_D$	83.84	87.13	74.01	80.88	53.44	81.54	62.18
		$+L_D + L_{uce}$	84.00	87.11	74.48	81.48	**53.79**	81.82	62.89
		$+L_D + L_{uce_cs}$	**84.33**	**87.52**	**74.90**	**82.09**	53.14	**82.26**	**63.29**
ResNet-50	1/8	baseline	80.86	83.71	71.06	82.26	44.77	79.50	58.74
		$+L_D$	81.74	83.84	73.08	82.60	58.00	80.35	62.02
		$+L_D + L_{uce}$	82.74	85.30	**73.76**	83.68	**64.37**	81.43	64.56
		$+L_D + L_{uce_cs}$	**83.03**	**86.33**	73.57	**83.72**	63.24	**81.82**	**64.68**
	1/4	baseline	83.80	87.05	74.05	81.82	58.21	81.69	63.66
		$+L_D$	85.09	88.28	75.11	82.69	66.93	82.93	66.80
		$+L_D + L_{uce}$	**85.73**	**89.12**	75.89	83.36	68.55	**83.68**	68.04
		$+L_D + L_{uce_cs}$	85.67	88.96	**76.11**	**83.48**	**68.69**	**83.68**	**68.29**

* the FCN with ResNet-50 is used as baseline and for all the SSL methods



(a) (b) (c) (d) (e)

Fig. 6. Comparison (whole image) between our method and other supervised learning methods on the Vaihingen test set. (a) Input images. (b) Ground truth. (c) Ours method (FCN). (d) DeeplabV3. (e) PSPNet.



(a) (b) (c) (d) (e)

Fig. 7. Comparison (whole image) between our method and other supervised learning methods on the Potsdam test set. (a) Input images. (b) Ground truth. (c) Ours method (FCN). (d) DeeplabV3. (e) PSPNet.

buildings, and low vegetation. Since different labeling ratios correspond to the weight distribution between supervised learning and unsupervised learning in semisupervised learning, our improvement on the Potsdam dataset mainly performs in supervised learning instead of unsupervised learning. This shows that our method makes better improvement in less labeled data. Fig. 7 shows the whole image segmentation results of DeepLab_v3 [51], PSPNet [49], and our method (FCN) on Potsdam test set.

D. Ablation Study

We conducted ablation experiments on the Vaihingen dataset. We set up a total of four groups of experiments with ResNet-18 and ResNet-50 in 1/8 and 1/4 labeled data, respectively. In each group of experiments, we analyzes four models—adopting only supervised learning (*baseline*), adopting our discriminator training strategy ($+L_D$), adopting confidence-weighted unsupervised loss ($+L_D + L_{\text{uce}}$) and adopting consistency self-training ($+L_D + L_{\text{uce_cs}}$). Table VIII shows the performance of four groups of experiments.

1) Discriminator Training: The pixel-level comparison between prediction and ground truth (denoted as $+L_D$) yields 58.30% (8.95% ↑) of mean IoU in the first experiment and 62.18% (7.07% ↑) of mean IoU in the second experiment. And $+L_D$ yields 62.02% (3.28% ↑) of mean IoU in the third experiment and 66.80% (3.14% ↑) of mean IoU in the fourth experiment. This shows that even for the lightweight backbone is poor, our discriminator training strategy ($+L_D$) can effectively learn pixel-level information of the positive prediction in the unlabeled data. Observing the *baseline* in the first and third experiments, we found that the mean IoU increases by 9.39% when ResNet-50 replaces ResNet-18. The improvement brought by ResNet-50 is almost consistent with the improvement brought by $+L_D$ when using ResNet-18. This shows that $+L_D$ can bring a similar effect to the backbone improvement even if in less labeled data.

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2) Confidence-Weighted Self-Training: Compared with $+L_D$, the confidence-weighted unsupervised loss (denoted as $+L_D + L_{\text{uce}}$) improves by 0.34% of mean IoU in the first experiment, by 0.71% of mean IoU in the second experiment, by 2.54% of mean IoU in the third experiment, and by 1.24% of mean IoU in forth experiment. This shows that using the confidence of the discriminator as the weight of the unsupervised loss can mitigate the impact of some noise in pseudo-labels. However, using ResNet-50 as the backbone can bring more significant improvement than using ResNet-18 as the backbone. This is because better generators can help to train better discriminators thereby making more credible confidence assessment. The confidence assessment when using ResNet-50 as the backbone as the generator is more credible than using ResNet-18 as the backbone.

3) Consistency Self-Training: Compared the results with $+L_D$, the consistency constrained unsupervised loss (denoted as $+L_D + L_{\text{uce_cs}}$) improves by 1.99% of mean IoU in the first experiment, by 1.11% of mean IoU in the second experiment, by 2.66% of mean IoU in the third experiment and 1.49% of mean IoU in the fourth experiment. The improvement in the first and second experiments is more obvious than in the third and fourth experiments. This is because the decision-level fusion of the predictions with all perturbations can keep the information of the positive prediction area which maintains consistency under all perturbations to make a more credible confidence assessment.

V. Conclusion

In view of the high cost of pixel-level annotation of high-resolution RS images and the large demand of data for supervised learning, we propose semisupervised learning with consistency self-training framework. In this framework, our discriminator assesses accurately the confidence of predictions, which reweights unsupervised loss to balance the improvement of each area in pseudo-labels. We measure the relative entropy by the cross entropy loss to maintain prediction consistency with all perturbations. We proved the effectiveness of our method in the iSAID datasets and the ISPRS datasets. We also analyzed the problem of existing semisupervised methods based on GAN and consistency regularization. Experiments have shown that our method combines the advantages of these two strategies to solve the difficulties of accurate pseudo-label identification and prediction consistency constraint.

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